

Topic 09: Exact Differential Equations & Integrating Multipliers

Professor Leonard Master Series · Lessons 28 to 30

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Gradient Vector Field

Professor Leonard bridges differential equations with Multivariable Calculus:

"Remember Clairaut's Theorem on mixed partials: $f_{xy} = f_{yx}$? An exact differential equation $M(x, y)dx + N(x, y)dy = 0$ is just the total differential $d\psi = 0$ of some hidden 3D potential surface $\psi(x, y) = C$. The solution curves are literally the contour lines (level curves) of that surface! If $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the vector field is conservative, and we can find $\psi(x, y)$ by integrating!"

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 9.1: Standard Exact Differential Equation

Statement: Solve $(2xy + 3)dx + (x^2 - 1)dy = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Test for Exactness:**

$$M(x, y) = 2xy + 3 \implies \frac{\partial M}{\partial y} = 2x$$

$$N(x, y) = x^2 - 1 \implies \frac{\partial N}{\partial x} = 2x$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \implies \text{The equation is EXACT!}$$

- **Step 2: Integrate $M(x, y)$ with respect to x :**

$$\psi(x, y) = \int (2xy + 3)dx = x^2y + 3x + h(y)$$

(Notice $h(y)$ replaces the constant $+C$ because y was held constant!)

- **Step 3: Differentiate $\psi(x, y)$ with respect to y and equate to $N(x, y)$:**

$$\frac{\partial \psi}{\partial y} = x^2 + h'(y)$$

Set equal to $N(x, y) = x^2 - 1$:

$$x^2 + h'(y) = x^2 - 1 \implies h'(y) = -1$$

- **Step 4: Integrate $h'(y)$:**

$$h(y) = -y$$

- **Step 5: Write the Level Curve Solution $\psi(x, y) = C$:**

$$x^2 y + 3x - y = C$$

Explicit solution: $y(x^2 - 1) = C - 3x \implies y(x) = \frac{C-3x}{x^2-1}$.

Problem 9.2: Trigonometric Exact Equation

Statement: Solve $(\cos y + y \cos x)dx + (\sin x - x \sin y)dy = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Exactness Test:**
 - $M_y = -\sin y + \cos x$.
 - $N_x = \cos x - \sin y$. $M_y = N_x \implies$ **Exact!**
- **Step 2: Integrate $M(x, y)$ with respect to x :**

$$\psi(x, y) = \int (\cos y + y \cos x)dx = x \cos y + y \sin x + h(y)$$

- **Step 3: Match to $N(x, y)$:**

$$\frac{\partial \psi}{\partial y} = -x \sin y + \sin x + h'(y)$$

$$(\sin x - x \sin y) + h'(y) = \sin x - x \sin y \implies h'(y) = 0 \implies h(y) = 0$$

- **Step 4: General Solution:**

$$x \cos y + y \sin x = C$$

Problem 9.3: Non-Exact Equation with Integrating Factor $\mu(x)$

Statement: Solve $(3xy + y^2)dx + (x^2 + xy)dy = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Test Exactness:** $M_y = 3x + 2y$, $N_x = 2x + y$. $M_y \neq N_x \implies$ **Not Exact!**
- **Step 2: Check Professor Leonard's Multiplier Formula for $\mu(x)$:**

$$\frac{M_y - N_x}{N} = \frac{(3x + 2y) - (2x + y)}{x^2 + xy} = \frac{x + y}{x(x + y)} = \frac{1}{x}$$

Depends

only on x

!

- **Step 3: Compute $\mu(x)$:**

$$\mu(x) = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

- **Step 4: Multiply the entire ODE by x :**

$$(3x^2y + xy^2)dx + (x^3 + x^2y)dy = 0$$

Verify new exactness

:

- New $M_y = 3x^2 + 2xy$.
- New $N_x = 3x^2 + 2xy$. Exact!
- **Step 5: Integrate:**

$$\psi(x, y) = \int (3x^2y + xy^2)dx = x^3y + \frac{1}{2}x^2y^2 + h(y)$$

$$\frac{\partial \psi}{\partial y} = x^3 + x^2y + h'(y) = x^3 + x^2y \implies h'(y) = 0$$

- **Step 6: Final Solution:**

$$x^3y + \frac{1}{2}x^2y^2 = C$$

Problem 9.4: Non-Exact Equation with Integrating Factor $\mu(y)$

Statement: Solve $(y^2 + 2xy)dx - x^2dy = 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Test Exactness:** $M_y = 2y + 2x$, $N_x = -2x$. Not exact.
- **Step 2: Check $\frac{N_x - M_y}{M}$:**

$$\frac{N_x - M_y}{M} = \frac{-2x - (2y + 2x)}{y^2 + 2xy} = \frac{-4x - 2y}{y(y + 2x)} = \frac{-2(2x + y)}{y(2x + y)} = -\frac{2}{y}$$

Depends

only on y

!

- **Step 3: Compute $\mu(y)$:**

$$\mu(y) = e^{\int -\frac{2}{y} dy} = e^{-2 \ln y} = y^{-2} = \frac{1}{y^2}$$

- **Step 4: Multiply by $\mu(y) = 1/y^2$:**

$$\left(1 + \frac{2x}{y}\right) dx - \frac{x^2}{y^2} dy = 0$$

- **Step 5: Integrate:**

$$\psi(x, y) = \int \left(1 + \frac{2x}{y}\right) dx = x + \frac{x^2}{y} + h(y)$$

$$\frac{\partial \psi}{\partial y} = -\frac{x^2}{y^2} + h'(y) = -\frac{x^2}{y^2} \implies h'(y) = 0$$

- **Step 6: General Solution:**

$$x + \frac{x^2}{y} = C \implies y(x) = \frac{x^2}{C - x}$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting Negative Sign in $N(x, y)$:** If the equation is $Mdx - Ndy = 0$, $N(x, y)$ includes the negative sign!
- **Trap 2: Mixing up $\mu(x)$ vs $\mu(y)$ Formulas:**
 - For $\mu(x)$: use $\frac{M_y - N_x}{N}$.
 - For $\mu(y)$: use $\frac{N_x - M_y}{M}$. Notice the numerator signs flip!